

Homework 1 Group I

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Question 1

Let $s_i = 1 - 2y_i$. For $\lambda > 0$, the objective function is

$$f_\lambda(\theta) = \underbrace{\sum_{i=1}^m \phi(s_i \langle \tilde{x}_i, \theta \rangle)}_{f(\theta)} + \frac{\lambda}{2} \|\theta\|^2, \quad \theta \in E.$$

We first show that f is convex. The smoothed hinge loss ϕ satisfies

$$\phi'(z) = \begin{cases} 0, & z \leq -1, \\ 1 + z, & -1 < z < 0, \\ 1, & z \geq 0. \end{cases}$$

The values and piecewise derivatives agree at $z = -1$ and $z = 0$. Thus ϕ is continuously differentiable. Its derivative is nondecreasing, so for any $a, b \in \mathbb{R}$,

$$\phi(b) - \phi(a) - \phi'(a)(b - a) = \int_a^b (\phi'(u) - \phi'(a)) du \geq 0.$$

(The inequality holds for either ordering of a and b .) By Theorem 4.21 of the lecture notes, ϕ is convex.

For any $\theta, \eta \in E$ and $t \in [0, 1]$, linearity of the inner product and convexity of ϕ give

$$\begin{aligned} \phi(s_i \langle \tilde{x}_i, (1-t)\theta + t\eta \rangle) &= \phi((1-t)s_i \langle \tilde{x}_i, \theta \rangle + ts_i \langle \tilde{x}_i, \eta \rangle) \\ &\leq (1-t)\phi(s_i \langle \tilde{x}_i, \theta \rangle) + t\phi(s_i \langle \tilde{x}_i, \eta \rangle). \end{aligned}$$

Summing over i yields $f((1-t)\theta + t\eta) \leq (1-t)f(\theta) + tf(\eta)$; hence f is convex.

Now

$$f_\lambda(\theta) - \frac{\lambda}{2} \|\theta\|^2 = f(\theta)$$

is convex. Since $\lambda > 0$, Definition 4.5 of the lecture notes therefore implies that f_λ is λ -strongly convex. It is real valued on the whole Euclidean space E , so Theorem 4.28 applies: f_λ has a unique global minimizer θ^* .

Moreover, f_λ is convex and differentiable on E . Each loss term is a composition of the continuously differentiable function ϕ with a linear map, and the quadratic term is differentiable. By Corollary 4.23 of the lecture notes, a point is a global minimizer if and only if its gradient vanishes. So, what we can say about the critical points is that

$$\{\theta \in E : \nabla f_\lambda(\theta) = 0\} = \{\theta^*\}.$$

Question 2

We now set $\lambda = 0.005$. Write $g_i(\theta) = s_i \langle \tilde{x}_i, \theta \rangle$. Each $g_i : E \rightarrow \mathbb{R}$ is linear, hence differentiable, and ϕ is continuously differentiable as shown in Question 1. Thus $f = \sum_{i=1}^m \phi \circ g_i$ is differentiable. By additivity of the differential and the chain rule, for any $h \in E$,

$$Df(\theta)[h] = \sum_{i=1}^m \phi'(g_i(\theta)) Dg_i(\theta)[h].$$

Using the limit expression for the differential of g_i , we obtain

$$Dg_i(\theta)[h] = \lim_{t \rightarrow 0} \frac{g_i(\theta + th) - g_i(\theta)}{t} = \lim_{t \rightarrow 0} \frac{ts_i \langle \tilde{x}_i, h \rangle}{t} = s_i \langle \tilde{x}_i, h \rangle.$$

Substituting and using bilinearity of the inner product gives

$$\begin{aligned} Df(\theta)[h] &= \sum_{i=1}^m s_i \phi'(s_i \langle \tilde{x}_i, \theta \rangle) \langle \tilde{x}_i, h \rangle \\ &= \left\langle \sum_{i=1}^m s_i \phi'(s_i \langle \tilde{x}_i, \theta \rangle) \tilde{x}_i, h \right\rangle. \end{aligned}$$

The quadratic term $r(\theta) = \frac{\lambda}{2} \|\theta\|^2$ is also differentiable. Indeed,

$$r(\theta + h) - r(\theta) - \lambda \langle \theta, h \rangle = \frac{\lambda}{2} \|h\|^2 = o(\|h\|),$$

so $Dr(\theta)[h] = \lambda \langle \theta, h \rangle$. Consequently, $f_\lambda = f + r$ is differentiable and

$$Df_\lambda(\theta)[h] = \left\langle \sum_{i=1}^m s_i \phi'(s_i \langle \tilde{x}_i, \theta \rangle) \tilde{x}_i + \lambda \theta, h \right\rangle.$$

By Definition 2.18 of the lecture notes, $Df_\lambda(\theta)[h] = \langle \nabla f_\lambda(\theta), h \rangle$. Since this equality holds for every $h \in E$, we conclude that

$$\nabla f_\lambda(\theta) = \sum_{i=1}^m s_i \phi'(s_i \langle \tilde{x}_i, \theta \rangle) \tilde{x}_i + \lambda \theta.$$

Question 4

Given that ∇f_λ is L -Lipschitz continuous with constant $L = \sigma_{\max}(X)^2 + \lambda$ (where $\sigma_{\max}(X)$ is computed numerically), we select the constant step-size $\alpha = \frac{1}{L} = \frac{1}{\sigma_{\max}(X)^2 + \lambda}$ as a consequence of Theorems 3.3 and 3.4. .

Question 6

Consider the theoretically infinite sequence generated by gradient descent with a fixed step size $\alpha > 0$:

$$\theta_{k+1} = \theta_k - \alpha \nabla f_\lambda(\theta_k), \quad k = 0, 1, 2, \dots,$$

starting from an arbitrary point $\theta_0 \in E$.

We check whether Theorem 4.31 of the lecture notes applies.

From Question 1, f_λ is λ -strongly convex. Indeed,

$$f_\lambda(\theta) - \frac{\lambda}{2} \|\theta\|^2 = f(\theta)$$

is convex, so, by Definition 4.5 of the lecture notes, f_λ is λ -strongly convex. Hence, in the notation of Theorem 4.31, we may take

$$\mu = \lambda.$$

Since $\lambda > 0$, f_λ has a unique global minimizer, which we denote by θ^* .

Furthermore, let

$$X = [\tilde{x}_1 \quad \dots \quad \tilde{x}_m].$$

As stated in the homework, ∇f_λ is Lipschitz continuous with Lipschitz constant

$$L_0 = \sigma_{\max}(X)^2 + \lambda,$$

where $\sigma_{\max}(X)$ denotes the largest singular value of X . Thus, for all $\theta, \eta \in E$,

$$\|\nabla f_\lambda(\theta) - \nabla f_\lambda(\eta)\| \leq L_0 \|\theta - \eta\|.$$

Therefore all the assumptions of Theorem 4.31 are satisfied with

$$\mu = \lambda, \quad L = L_0 = \sigma_{\max}(X)^2 + \lambda.$$

For the step size prescribed by Theorem 4.31,

$$\alpha = \frac{1}{L_0} = \frac{1}{\sigma_{\max}(X)^2 + \lambda},$$

gradient descent satisfies, for every $k \geq 0$,

$$f_\lambda(\theta_k) - f_\lambda(\theta^*) \leq \left(1 - \frac{\lambda}{\sigma_{\max}(X)^2 + \lambda}\right)^k (f_\lambda(\theta_0) - f_\lambda(\theta^*)).$$

Hence $f_\lambda(\theta_k)$ converges globally and at least linearly to the optimal value $f_\lambda(\theta^*)$.

More generally, Theorem 4.31 can be applied to every fixed step size

$$0 < \alpha \leq \frac{1}{\sigma_{\max}(X)^2 + \lambda}.$$

Indeed, if

$$0 < \alpha \leq \frac{1}{L_0},$$

then $1/\alpha \geq L_0$. Since a function whose gradient is L_0 -Lipschitz also has L -Lipschitz gradient for every $L \geq L_0$, we may apply Theorem 4.31 with

$$L = \frac{1}{\alpha}.$$

The resulting estimate is

$$f_\lambda(\theta_k) - f_\lambda(\theta^*) \leq (1 - \alpha\lambda)^k (f_\lambda(\theta_0) - f_\lambda(\theta^*)).$$

We can also obtain convergence of the iterates themselves. Since f_λ is λ -strongly convex and θ^* is its minimizer, we have $\nabla f_\lambda(\theta^*) = 0$. Hence strong convexity gives

$$f_\lambda(\theta) \geq f_\lambda(\theta^*) + \frac{\lambda}{2} \|\theta - \theta^*\|^2.$$

Therefore,

$$\|\theta_k - \theta^*\|^2 \leq \frac{2}{\lambda} (f_\lambda(\theta_k) - f_\lambda(\theta^*)),$$

and using the estimate above,

$$\|\theta_k - \theta^*\| \leq \sqrt{\frac{2}{\lambda} (f_\lambda(\theta_0) - f_\lambda(\theta^*))} (1 - \alpha\lambda)^{k/2}.$$

Thus the iterates converge globally and linearly to the unique minimizer θ^* .

For the value used in the homework,

$$\lambda = 0.005,$$

the admissible step sizes certified by Theorem 4.31 are therefore

$$0 < \alpha \leq \frac{1}{\sigma_{\max}(X)^2 + 0.005}.$$

In particular, the largest step size directly certified using the given Lipschitz constant is

$$\alpha = \frac{1}{\sigma_{\max}(X)^2 + 0.005}.$$

Application to our implementation

We now check whether the preceding convergence result applies to the gradient descent method used in our implementation.

We use the fixed step size

$$\alpha = \frac{1}{L}, \quad L = \sigma_{\max}(X)^2 + \lambda.$$

As verified above, f_λ is λ -strongly convex and ∇f_λ is L -Lipschitz continuous. Therefore, our implemented gradient descent iteration

$$\theta_{k+1} = \theta_k - \frac{1}{L} \nabla f_\lambda(\theta_k)$$

is precisely in the setting of Theorem 4.31 of the lecture notes. Hence the theoretical convergence result applies to our algorithm.

In particular, if

$$\kappa = \frac{L}{\lambda} = \frac{\sigma_{\max}(X)^2 + \lambda}{\lambda},$$

then Theorem 4.31 gives

$$f_\lambda(\theta_k) - f_\lambda(\theta^*) \leq \left(1 - \frac{1}{\kappa}\right)^k (f_\lambda(\theta_0) - f_\lambda(\theta^*)).$$

Moreover, Corollary 4.32 of the lecture notes gives the stronger conclusions

$$\|\theta_k - \theta^*\| \leq \sqrt{\frac{2}{\lambda} (f_\lambda(\theta_0) - f_\lambda(\theta^*))} \left(\sqrt{1 - \frac{1}{\kappa}}\right)^k$$

and

$$\|\nabla f_\lambda(\theta_k)\| \leq \sqrt{2L (f_\lambda(\theta_0) - f_\lambda(\theta^*))} \left(\sqrt{1 - \frac{1}{\kappa}}\right)^k.$$

Thus, theoretically, both the iterates and the norm of the gradient converge at least linearly, with

$$\|\nabla f_\lambda(\theta_k)\| \longrightarrow 0.$$

Comparison with the finite run

This qualitative prediction is consistent with the behavior observed in our finite numerical run. The plot of $\|\nabla f_\lambda(\theta_k)\|$ shows a clear decrease as the iteration number increases. The gradient norm decreases very rapidly during the first iterations and then continues to decrease more slowly, eventually reaching the prescribed stopping tolerance after approximately 2.7×10^4 iterations.

This agrees qualitatively with Corollary 4.32, which predicts that the gradient norm tends to zero at least linearly. In particular, the corollary provides a geometric upper bound of the form

$$\|\nabla f_\lambda(\theta_k)\| \leq Cq^k, \quad q = \sqrt{1 - \frac{1}{\kappa}} = \sqrt{1 - \frac{\lambda}{L}} < 1,$$

where

$$C = \sqrt{2L(f_\lambda(\theta_0) - f_\lambda(\theta^*))}.$$

It is important that this is an upper bound and not a statement that the gradient norm must decrease by exactly the same factor at every iteration. Consequently, the changes of slope and the relatively flat parts visible in the numerical plot do not contradict the theoretical result. The theorem gives a worst-case global convergence guarantee, whereas the actual convergence speed may vary substantially along the trajectory.

Finally, our numerical run is finite and is terminated once

$$\|\nabla f_\lambda(\theta_k)\|$$

reaches the prescribed stopping tolerance. Therefore, the experiment does not literally show convergence of the gradient norm to zero. Rather, it shows the gradient norm decreasing until the stopping criterion is met. This is consistent with the asymptotic prediction of Corollary 4.32 for the theoretically infinite sequence.

Question 7

The classification error is approximately 0.134% on the training set and 0.189% on the test set. The test error is slightly higher than the training error, as expected, but remains very small, indicating that the classifier generalizes well to unseen samples.